

Part 1

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$$a) \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{(n^2-3n)^2}{5n^3+n} \Rightarrow \frac{\infty}{\infty} \text{ so we should use L'Hospital}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{(n^2-3n)^2}{(5n^3+n)'} = \lim_{n \rightarrow \infty} \frac{2(n^2-3n) \cdot (2n-3)}{15n^2+1} \Rightarrow \lim_{n \rightarrow \infty} \underbrace{2}_{2} \cdot \lim_{n \rightarrow \infty} \underbrace{2n-3}_{\infty} \cdot \lim_{n \rightarrow \infty} \frac{n^2-3n}{15n^2+1}$$

$$\lim_{n \rightarrow \infty} \frac{n^2(1-\frac{3}{n})}{n^2(15+\frac{1}{n^2})} \Rightarrow \lim_{n \rightarrow \infty} \frac{1-\frac{3}{n}}{15+\frac{1}{n^2}} \rightarrow \text{it goes 0} \rightarrow \text{so } \lim_{n \rightarrow \infty} \frac{n^2-3n}{15n^2+1} = \frac{1}{15} \Rightarrow \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 2 \cdot \infty \cdot \frac{1}{15} = \frac{2}{15} \cdot \infty = \infty$$

So $f(n) = \sqrt{g(n)}$

$$b) \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^3}{\log_2 n^4} \Rightarrow \frac{\infty}{\infty} \text{ so we should use L'Hospital}$$

$$\lim_{n \rightarrow \infty} \frac{(n^3)'}{(\log_2 n^4)'} \Rightarrow \lim_{n \rightarrow \infty} \frac{3n^2}{\frac{4n^3}{n^4} \log_2 e} \Rightarrow \lim_{n \rightarrow \infty} \frac{3n^6}{4n^3 \log_2 e} = \lim_{n \rightarrow \infty} \frac{3}{4 \log_2 e} \cdot \lim_{n \rightarrow \infty} n^3$$

this is constant number. so

$$\lim_{n \rightarrow \infty} \frac{3}{4 \log_2 e} = \frac{3}{4 \log_2 e}, \lim_{n \rightarrow \infty} n^3 = \infty$$

$$\Rightarrow \frac{3}{4 \log_2 e} \cdot \infty = \infty \text{ so } f(n) = \sqrt{g(n)}$$

$$c) \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{5n \cdot \log_2 n}{n \cdot \log_2 5^n} \Rightarrow \frac{\infty}{\infty} \text{ so we should use L'Hospital.}$$

but before, let's simplify the functions, $\frac{5n}{n} = 5$ so $\lim_{n \rightarrow \infty} = \lim_{n \rightarrow \infty} \frac{5 \cdot \log_2 n}{\log_2 5^n} = \lim_{n \rightarrow \infty} 5 \cdot \lim_{n \rightarrow \infty} \frac{\log_2 n}{\log_2 5^n}$

$$\lim_{n \rightarrow \infty} 5 = 5$$

$$\lim_{n \rightarrow \infty} \frac{\log_2 n}{\log_2 5^n} = \frac{\infty}{\infty}, \text{ we use the L'Hospital's here}$$

$$\lim_{n \rightarrow \infty} \frac{(\log_2 n)'}{(\log_2 5^n)'} = \frac{\frac{1}{4n} \cdot \log_2 e}{\frac{5^n \cdot \ln e \cdot \log_2 e}{5^n}} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{4n} \cdot \frac{1}{\ln e} \Rightarrow \frac{1}{\ln e} \text{ is constant so}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{\ln e} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0 \text{ so } f(n) = O(g(n))$$

d) $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^n}{10^n} = \frac{\infty}{\infty}$ so we should use L'Hospital's rules

$\lim_{n \rightarrow \infty} \frac{(n^n)'}{(10^n)'} \Rightarrow n^n$ is a little bit complicated.

$\ln y = \ln n^n = n \cdot \ln n$
take the derivative of two side $\Rightarrow \frac{y'}{y} = \ln n + n \cdot \frac{1}{n}$
 $y' = n^n \cdot \ln n + n^n \cdot 1$

$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^n \cdot (\ln n + 1)}{10^n \cdot \ln 10} = \text{it is not possible to simplify.}$

(1) so $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \left(\frac{n}{10}\right)^n = \infty^{\infty} = \infty$, so $f(n) = \Omega(g(n))$

e) $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \frac{8n \cdot \sqrt[n]{2n}}{n \cdot \sqrt[n]{n}}$ $\Rightarrow$ let simplify the given statement

$\Rightarrow \sqrt[n]{2n} = \sqrt[n]{(2n)^3} = \sqrt[n]{8n^3} \Rightarrow \frac{8n}{n} = 8 \Rightarrow \lim_{n \rightarrow \infty} \frac{8 \cdot \sqrt[n]{8n^3}}{\sqrt[n]{n^5}} \Rightarrow \lim_{n \rightarrow \infty} 8 \cdot \lim_{n \rightarrow \infty} \sqrt[n]{\frac{8}{n^2}}$

$\Rightarrow \sqrt[n]{n} = \sqrt[n]{(n)^5}$

$\Rightarrow \lim_{n \rightarrow \infty} 8 = 8, \lim_{n \rightarrow \infty} \sqrt[n]{\frac{8}{n^2}} = 0 \Rightarrow 8 \cdot 0 = 0$ so $f(n) = O(g(n))$

Part 2

a) `for (int i=0; i < str_array.length; i++)
 str_array[i] = "";`

→ we know that `str_array[i] = ""` has $O(1)$ complexity but this statement is in for loop so it will do this n times

$$\Rightarrow O(n) \cdot O(1) = O(n)$$

b) `for (int i=0; i < str_array.length; i++)
 method A (str_array);
for (int j=0; j < str_array.length; j++)
 System.out.println(str_array[j]);`

Firstly, we should know that these

for loops are not inside each other they are separate.

Let's examine for first loop. this loop calls method a n times. method called i prints the array's element one by one. also includes another for loop so $n \times n$ times prints so complexity is $O(n \times n) = O(n^2)$

Let's examine second for loop. it just prints array element one by one.

Print statement's complexity is 1 but it is inside for loop so it is called n times. complexity is $n \times 1 = n = O(n)$

$$\text{Total complexity is } O(n^2) + O(n) = O(n^2 + n) = O(n^2)$$

c) `for (int i=0; i < str_array.length; i++)
 for (int j=0; j < str_array.length; j++)
 method B (str_array)`

we should know that these for loops are inside each other.

so method B (str_array) complexity = $n \cdot n$: method B (str_array)

$$\text{method B (str_array) complexity is } n^2 \Rightarrow \text{so } n^2 \cdot n^2 = n^4 = O(n^4)$$

d) `for (int i=0; i < str_array.length; i++) {
 System.out.println(str_array[i]);
 str_array[i--] = "";
}`

→ print statement is $O(1)$

decreasing the value of i ($i--$) is $O(1)$

assigning the array `""` is $O(1)$

but there is a problem here.

at the same time, we are trying to decrease i values else increase.

this is problem. the code will never finish so complexity is infinite. i will be 0 and -1 regularly. and the function will print "" infinite times. complexity is $O(\infty)$

e) `for (int i=0; i < str_array.length; i++) {
 if (str_array[i] == "")
 break;
}`

→ the complexity of break is $O(1)$

it does this n times

$$\text{so } T(n) = n \cdot O(1) = O(n)$$

also we cannot know when if statement occurs so we should think the worst case. maybe never if block will occur so $O(n)$ will be our time complexity.

Part 3

a) First, initialize the array with the appropriate values.
• we are assuming that array is ascending order and we are trying to find max-min value of array element. So first element is $A[0]$ which is minimum number of array. Besides, the maximum element is $A[n-1]$
Set $\min = \text{array}[0]$ $\rightarrow$ complexity = $O(1)$
Set $\max = \text{array}[\text{array.length}-1]$ $\rightarrow$ complexity = $O(1)$
Set $\text{value} = \max - \min$ $\rightarrow$ complexity = $O(1)$ $\rightarrow$ so total complexity = $O(1)$
maybe if statement can be added if $(n > 0)$ but in the given question, n is positive.
return;

b) First, initialize the array with the appropriate values.
we are assuming that array is not in ascending order.
Set $\min = \text{array}[0]$ $\rightarrow$ the complexity = $O(1)$ // this is not important.
Set $\max = \text{array}[0]$ $\rightarrow$ the complexity = $O(1)$

For $i = 0$ to n (n is not valid)

if $A[i]$ is smaller or equal than $\min$
assign $\min = A[i]$ // complexity = $O(1)$

end if statement.

if $A[i]$ is greater or equal than $\max$

assign $\max = A[i]$ // complexity = $O(1)$
end if

but it does this process n times. so complexity is $O(1) \cdot O(n) = O(n)$
then set $\text{value} = \max - \min$

return;

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